

3.6 Chain Rule Worksheet

1. Using Leibniz notation $\left(i.e., \frac{dy}{dx}, \frac{dP}{dt}, etc.\right)$, find the derivatives for each of the following functions.

a) $w = 10(q+1)^8 e^q + e^{\cos\left(\frac{\pi}{12}\right)}$

$$\frac{dw}{dq} = 10(q+1)^8 e^q + 80(q+1)^7 e^q = 10e^q (q+1)^7 [(q+1)+8] = 10e^q (q+1)^7 [q+9]$$

b) $x = 10y^3 \sin(2y) + \ln\left(\frac{5}{e^2}\right)$

$$\frac{dx}{dy} = 30y^2 \sin(2y) + 10y^3 \cos(2y) 2 = 10y^2 [3 \sin(2y) + 2y \cos(2y)]$$

c) $q = \frac{4 \sin(t)}{t^2 + 3}$

$$\frac{dq}{dt} = \frac{(t^2 + 3) 4 \cos(t) - 4 \sin(t)(2t)}{(t^2 + 3)^2}$$

d) $k = \sec(3t) + \sin\left(\frac{1}{e}\right)$

$$\frac{dk}{dt} = \sec(3t) \tan(3t)(3) = 3 \sec(3t) \tan(3t)$$

e) $y = \frac{(3x^2 - 1)^2}{\sin(e^x)}$

We *can* begin by rewriting the function as $y = (3x^2 - 1)^2 \csc(e^x)$.

$$\begin{aligned} \text{Then we calculate: } \frac{dy}{dx} &= 2(3x^2 - 1)(6x) \csc(e^x) + (3x^2 - 1)^2 [-\csc(e^x) \cot(e^x) e^x] \\ &= 12x(3x^2 - 1) \csc(e^x) - e^x (3x^2 - 1) \csc(e^x) \cot(e^x) \end{aligned}$$

$$f) \quad g = \cos(\tan(x^8))$$

$$\frac{dg}{dx} = -\sin(\tan(x^8)) \cdot \sec^2(x^8) \cdot (8x^7)$$

$$g) \quad v = e^{13t} \cot(25t)$$

$$\frac{dv}{dt} = e^{13t} (13) \cot(25t) - e^{13t} \csc^2(25t) (25)$$

$$h) \quad w = \sin\left(\frac{v^{12}}{e^v}\right)$$

$$\frac{dw}{dv} = \cos\left(\frac{v^{12}}{e^v}\right) \cdot \frac{e^v (12v^{11}) - e^v v^{12}}{(e^v)^2} = \cos\left(\frac{v^{12}}{e^v}\right) \cdot \frac{12v^{11} - v^{12}}{e^v}$$

$$2. \quad \text{Let } h(x) = \sqrt{4+3f(x)}. \text{ If } f(1) = 7 \text{ \& } f'(1) = 4. \text{ Find } h'(1).$$

$$h'(x) = \frac{1}{2} (4+3f(x))^{-1/2} \cdot 3f'(x) = \frac{3f'(x)}{2\sqrt{4+3f(x)}}$$

$$h'(1) = \frac{3f'(1)}{2\sqrt{4+3f(1)}} = \frac{3(4)}{2\sqrt{4+3(7)}} = \frac{12}{2\sqrt{25}} = \frac{12}{10} = \frac{6}{5}$$

3. Differentiate $y = \cot(x^4 + 8x)$.

$$y' = -\csc^2(x^4 + 8x) \cdot (4x^3 + 8)$$

4. Differentiate $y = [4\sin(x) + 2\cos(x)]^{3/2}$.

$$y' = \frac{3}{2} [4\sin(x) + 2\cos(x)]^{1/2} \cdot [4\cos(x) - 2\sin(x)]$$

5. Differentiate $y = \sqrt{\sin(e^{t^3+4t})}$.

$$y = \sqrt{\sin(e^{t^3+4t})} = [\sin(e^{t^3+4t})]^{1/2}$$

$$y' = \frac{1}{2} [\sin(e^{t^3+4t})]^{-1/2} \cdot \cos(e^{t^3+4t}) \cdot (e^{t^3+4t}) \cdot (3t^2 + 4)$$

6. Differentiate $y = \left[\frac{x^2}{x^5 + 1} \right]^4$.

$$y' = 4 \left[\frac{x^2}{x^5 + 1} \right]^3 \cdot \frac{(x^5 + 1)(2x) - (5x^4)(x^2)}{(x^5 + 1)^2} = 4 \left[\frac{x^2}{x^5 + 1} \right]^3 \cdot \frac{2x(x^5 + 1) - 5x^6}{(x^5 + 1)^2}$$

7. Differentiate $y = \sin(\cos(\tan(x)))$.

$$\begin{aligned}y' &= \cos(\cos(\tan(x))) \cdot [-\sin(\tan(x))] \cdot \sec^2(x) \\&= -\cos(\cos(\tan(x))) \sin(\tan(x)) \sec^2(x)\end{aligned}$$

8. Determine the equation of the line which is tangent to the curve $y = 4 \tan\left(\frac{x}{4}\right) + 5 \cos(x) + \pi$ at $x = \pi$.

$$y(\pi) = 4 \tan\left(\frac{\pi}{4}\right) + 5 \cos(\pi) + \pi = 4(1) + 5(-1) + \pi = \pi - 1$$

We are working with the point: $(\pi, \pi - 1)$

We calculate the derivative to be:

$$y' = 4 \sec^2\left(\frac{x}{4}\right) \left(\frac{1}{4}\right) - 5 \sin(x) = \sec^2\left(\frac{x}{4}\right) - 5 \sin(x)$$

$$y'(\pi) = \sec^2\left(\frac{\pi}{4}\right) - 5 \sin(\pi) = (\sqrt{2})^2 - 5(0) = 2$$

Therefore, the equation of the tangent line is:

$$y - (\pi - 1) = 2(x - \pi) \rightarrow y = 2x - 2\pi + (\pi - 1) \rightarrow y = 2x - (\pi + 1)$$

9. One of the points on the graph of $f(x) = e^{-9x}$ has the property that the line tangent to the curve at that point is perpendicular to the line $y = 9x + 3$. Find and fully simplify the y-coordinate for this point.

If the line tangent to the curve is perpendicular to the line $y = 9x + 3$, then that line has a slope of $-\frac{1}{9}$.

We calculate that: $f'(x) = -9e^{-9x}$.

We then need to find the value of x for which $-\frac{1}{9} = -9e^{-9x}$.

$$-\frac{1}{9} = -9e^{-9x} \rightarrow \frac{1}{81} = e^{-9x} \rightarrow \ln\left(\frac{1}{81}\right) = \ln(e^{-9x}) \rightarrow -\ln(81) = -9x \rightarrow x = \frac{\ln(81)}{9}$$

Therefore, the y-coordinate of this point must be given by: $f\left(\frac{\ln(81)}{9}\right) = e^{-9 \frac{\ln(81)}{9}} = e^{-\ln(81)} = e^{\ln\left(\frac{1}{81}\right)} = \frac{1}{81}$

10. Find the equation of the line which is tangent to the curve $y = 2e^x - 6$ at its x -intercept.

The x -intercept occurs when $y = 0$.

$$0 = 2e^x - 6 \rightarrow 3 = e^x \rightarrow x = \ln(3)$$

So the x -intercept is $(\ln(3), 0)$.

We then calculate the derivative and find that:

$$y' = 2e^x$$

$$y'(\ln(3)) = 2e^{\ln(3)} = 2(3) = 6$$

Therefore, the equation of the tangent line is:

$$y - 0 = 6(x - \ln(3))$$

$$y = 6x - 6\ln(3)$$